PROBLEM STATEMENT 01

RADIO TALK

Pit Wall Telemetry

The Silent Co-Driver

Reading driver stress from radio calls
and putting it beside the pace data engineers already watch.

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THE PROBLEM

The pit wall is flooded with numbers. The warning is human.

Radio carries information that telemetry cannot — but it is easy to miss under race-day load.

> “The driver sounds different.
> Nobody has time to listen closely.

Stress, frustration or fatigue cues can sit inside a 10-second radio call while engineers are simultaneously watching pace, tyres, track status and strategy.

01 **SIGNAL LOST**
Tone is buried under radio noise and constant comms.

02 **CONTEXT LOST**
A concerning call is separated from the lap it belongs to.

03 **ATTENTION LOST**
Humans cannot continuously listen + analyse telemetry.

THE GAP **Telemetry explains WHAT happened. Radio can hint at WHY the driver is reacting.**

THE SOLUTION

One radio clip becomes an operator-ready story.

Upload or play a clip. We turn speech + tone + lap context into one compact pit-wall view.

01
RADIO
Upload / play / record

02
SPEECH
Transcript

03
STATE
Mood + fatigue cues

04
PACE
Lap-time context

OUTPUT “Frustrated” + transcript + Lap 42 + pace delta + race context → engineer decides

The system surfaces evidence. It does not diagnose the driver or claim causation.

AI PIPELINE

Three independent signals. One conservative interpretation.

The pipeline is designed to degrade gracefully when one model is uncertain or unavailable.

- A — **WHISPER LARGE V3 (Hugging Face)**
  Speech → text
- B — **WAV2VEC2 XLS-R (Hugging Face)**
  Voice-tone emotion
- C — **DISTILROBERTA (Hugging Face)**
  Transcript sentiment
- D — **FUSION + RULES**
  Mood + confidence + source

CONSERVATIVE FALLBACK

VOICE
reliable acoustic signal

TRANSCRIPT
sentiment from words

IF RADIO NOISE BREAKS THE ACOUSTIC SIGNAL

→ use transcript-derived estimate and label the source transparently.

No black-box “100% stress” verdict.

THE DASHBOARD

Built for the engineer who has 30 seconds, not 30 minutes.

The interface puts the radio, model output and pace context on the same screen.

PIT WALL / SESSION REVIEW

RACE · LAP 42

TEAM RADIO

HAM · Ferrari

FRUSTRATED

“I got no power man. Come on guys.”

TRANSCRIPT

confidence · source: combined

DRIVER STATE

FRUSTRATED

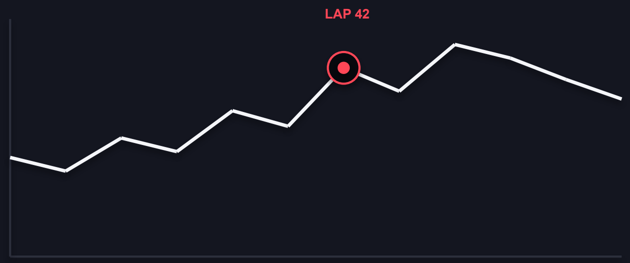
signal confidence

FATIGUE CUE

NO SIGNAL

LAP TIMES & RADIO MOOD

ONE POINT = ONE LAP · RING = SELECTED RADIO



+0.85s vs median

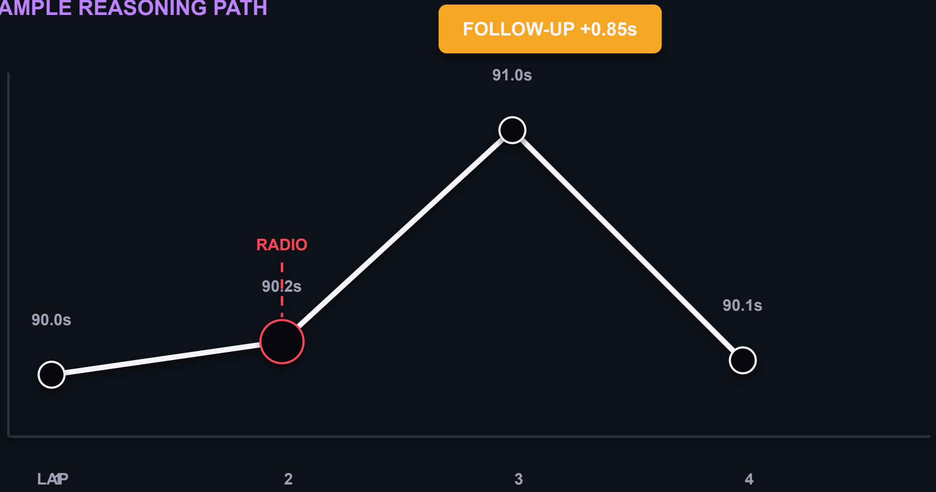
context checked

CORRELATION, NOT CAUSATION

When a concerning radio is followed by slower pace, we flag the lap *then* check race context.

The dashboard explicitly separates driver-state signals from explanations such as pits, safety car/VSC and rain.

EXAMPLE REASONING PATH



01 CONCERNING RADIO

Frustrated signal at lap 2

02 PACE CHANGE

Lap 3 is 0.85s slower than baseline

03 CONTEXT CHECK

Pit / SC / VSC / rain can explain pace.
Engineer validates before acting.

AI INSIGHT “Frustration at Lap 2 is followed by a +0.85s pace drop on Lap 3 → investigate before attributing the slowdown to driver state.”

RACE-DAY ROBUSTNESS

The system is built around uncertainty → not hidden behind it.

Every service can report its own status, and the UI tells the operator where the signal came from.

✓ **MODEL FAILURE**

If ASR, audio emotion or text sentiment fails, the other services can still complete.

✓ **NOISY RADIO**

Acoustic tone can fall back to transcript-derived sentiment instead of fabricating confidence.

✓ **FATIGUE CUES**

Only explicit tiredness/focus wording is screened; it is not a medical assessment.

✓ **RACE CONTEXT**

Pit laps, safety-car/VSC and rain are considered before implying driver-state effects.

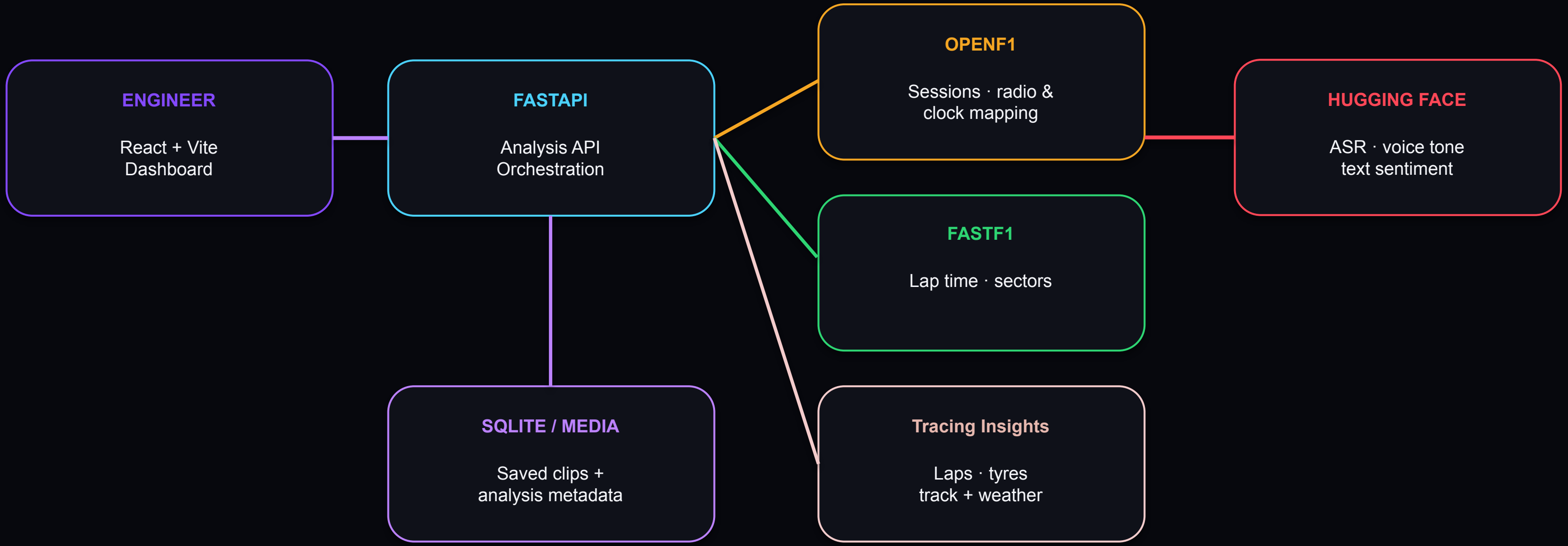
DESIGN PRINCIPLE

Surface uncertainty → preserve evidence → keep the engineer in the loop.

UNDER THE HOOD

A lightweight web stack connects public race data, AI inference and a local review store.

The prototype is deployable as a single Docker Compose stack.



DOCKER COMPOSE Frontend + FastAPI backend + local persistence · HF token supplied at runtime

THE TAKEAWAY

We are not replacing the race engineer.

We are giving them one extra set of ears — attached to the numbers they already trust.

LISTEN.

UNDERSTAND.

CONTEXTUALISE.

ACT.

PIT WALL TELEMETRY

Radio audio	→	Driver-state signal
Lap context	→	Engineer decision

Faster awareness.
Better context.
Human remains in control.

THE SILENT CO-DRIVER